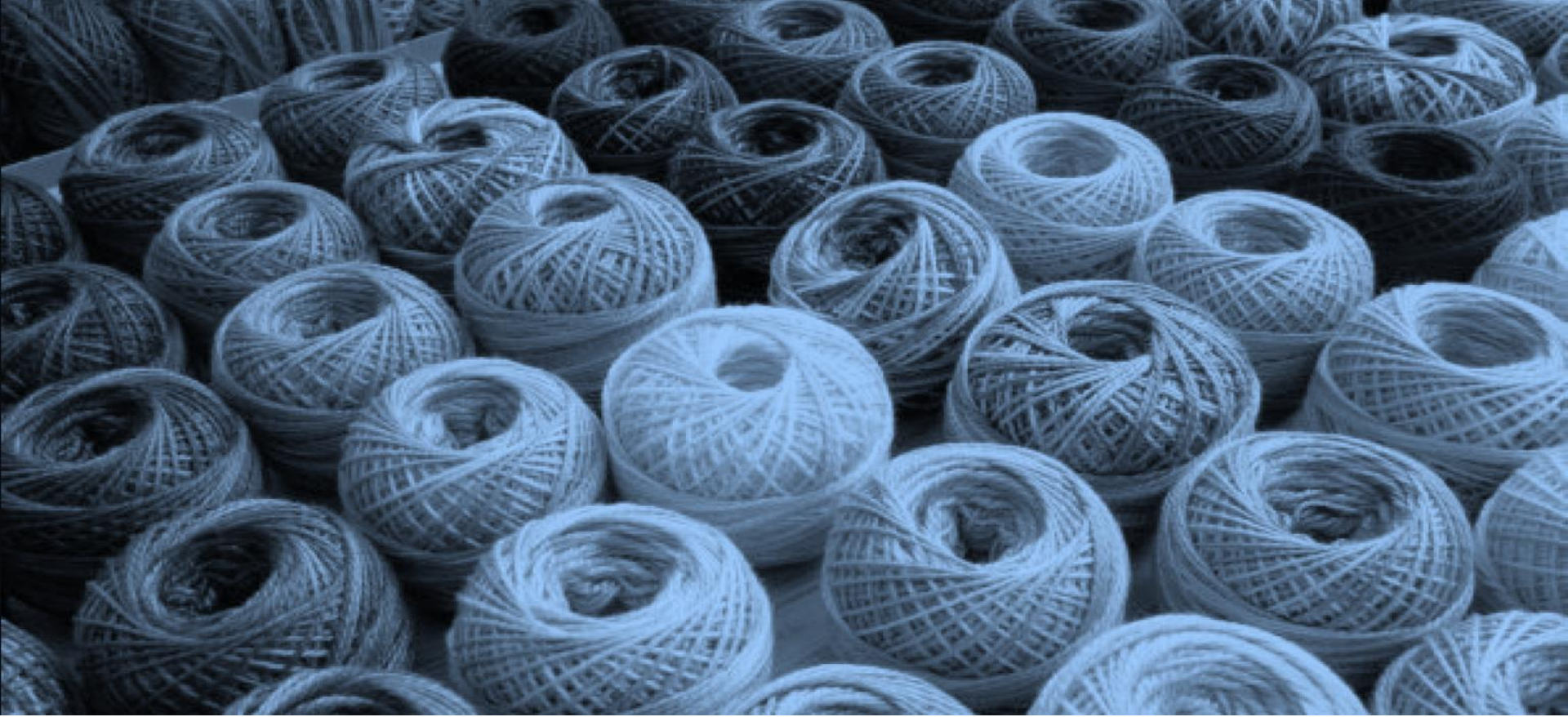

Java Programming

CPT111 – Lab 5

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CPT111 Java Programming

Lab 5

Static Methods 2 and String

Lab Exercise 5.1 – War and Peace

- In this lab, you will learn problem solving using methods, chars, and strings, by creating test cases first

QUIZ

Lab Exercise 5.1 War and Peace

- **Complete** the method that takes a string and returns an boolean:

Write a Java method that given a string `text`, returns `true` if the string "war" and "peace" appear the same number of times in `text`.

All input letters are in lowercase.

Test cases :
warAndPeace("there was never a good war, or a bad peace") → true
warAndPeace("war what is it good for") → false

Lab Exercise 5.1 – Activity 1: Create Test Cases

- First, **create** test cases for the following cases:
  - one occurrence of "war" and one occurrence of "peace" (the Given Test Case 1)
  - one "war" and zero "peace" (the Given Test Case 2)

  - zero "war" and one "peace"
  - multiple "war" and "peace", and the same number
  - multiple "war" and "peace", but different number
  - strings where "war" is immediately followed by "peace"
  - strings where "peace" is immediately followed by "war"
  - strings similar to above but also containing substrings of "war" and "peace"
  - empty string

Lab Exercise 5.1 – Activity 2: Solve the Problem

- Write your code and continuously refine it to pass more test cases
  - there can be multiple possible solutions
  - you could use indexOf() which you have learned during lecture
- Once you have passed all of your test cases, submit to Learning Mall
  - if your code fails, add more test cases and debug it!

	Test	Expected	Got	
✓	`System.out.println(warAndPeace("there was never a good war, or a bad peace"));`	true	true	✓
✓	`System.out.println(warAndPeace("war what is it good for"));`	false	false	✓

Passed all tests! ✓

Correct

Marks for this submission: 100.00/100.00.

Lab 5 Challenge – Sum of Proper Elements

- In the lab challenge, you will solve another problem using methods, chars, and strings:

QUIZ

Lab 5 Challenge - Remove All Leftmost Substring

- **Complete** the method that takes two strings: the *input* string and the *target* substring

Complete the method `removeLeftmostSubstring`.

The method removes the leftmost occurrence of target substring from input string,
repeatedly on the resulting string,
until all occurrences of target are removed.
It returns the resulting string after repeated removal.

You must use String methods in lecture notes.
You must NOT use StringBuilder, Java Util, or Regular Expression methods: append, matches, replaceAll.

Test cases:

```
removeLeftmostSubstring("abcd", "abc") → "d"

removeLeftmostSubstring("ababccd", "abc") → "abcd" → "d"
```

(two removals happened in the last test case)

Lab 5 Challenge – Task 1: Create Test Cases (50 marks)

- **Add** more test cases to the test class
  - try to include cases that cover all possible *output*: 0, 1, 2, and more **removals**,
  - when the previous removals *introduce* target substring,
  - and all kinds of combinations *input* strings and *target* substrings

Lab 5 Challenge – Task 2: Solve the Problem (50 marks)

- **Write** your code and continuously refine it to pass more test cases
- Once you have passed all of your test cases, submit to Learning Mall
  - if your code fails, add more test cases and debug it!
- **Show** to your Lab TA that you have passed the Learning Mall Quiz:

	Test	Expected	Got	
✔	`int[] nums = {1, 2, 6};` `System.out.println(sumOfProper(nums));`	9	9	✔
✔	`int[] nums = {10, 25};` `System.out.println(sumOfProper(nums));`	10	10	✔

Passed all tests! ✔

Correct

Marks for this submission: 100.00/100.00.

- **Briefly explain** to the TA you solution!

Thank you for your hard work!

- If you finish early, feel free to use the remaining time to work on other online exercises
  - use this opportunity to practice your new skills in testing your code!

- You may leave the lab once you have complete the Lab 5 Challenge and it has been ASSESSED by a TA !
  - Please note that the **Lab 5 Challenge** is **graded** and contributes to your CW2 marks